FUTURE HEALTH | A GLOBAL INITIATIVE BY ABU DHABI | H2S

# Future Health Digital Innovation Challenge

From signals to solutions.

Co-creator Microsoft

Co-creator inception42

Co-creator Mohamed bin Zayed University of Artificial Intelligence

Team Name: Silah · صلة
Team Size: 1
Problem Statement: Fertility · Challenge 03, Understanding and Reducing Inherited Disease Risk

Product page, demo and film

*A private companion that helps Abu Dhabi families act on inherited disease risk before the next pregnancy, at birth and across the family, without ever deciding for them.*

Team members*

# A solo entry, built end to end by one person

| S.No | Name | Country | Role | Organization | Age confirmation |
|---|---|---|---|---|---|
| 1 | Marc Donovici | France | Solo: product, engineering, research | Independent | Over 18: confirmed |

A solo entry. Every part of Silah, from the risk arithmetic and the cryptography to the Arabic and English interface and the simulated health-system side, was designed, built and tested by one person.

Team Capability Summary*

# Offline clinical AI, built and audited by the same person

I audit AI systems for a living and build offline clinical AI.

I lead AI for the Internal Audit function at Crédit Mutuel Alliance Fédérale, where I also audit information systems against GDPR and the EU AI Act. One mission covered the group's own AI architecture and its governance and security controls.

In clinical AI, FieldScreen AI, an offline tuberculosis screening tool for community health workers, placed 3rd of 850+ teams in Google's MedGemma Impact Challenge 2026.

For Silah I wrote the risk kernel, the verifier, the cryptography, the phone app and the simulated health-system side, and ran the model evaluation.

## 3rd of 850+

Google MedGemma Impact Challenge 2026, offline TB screening

## Finalist, 1 of 6

European Patent Office CodeFest 2026, explainable ML

## Track winner

EdgeAgent track, Global AI Hackathon with Qwen Cloud

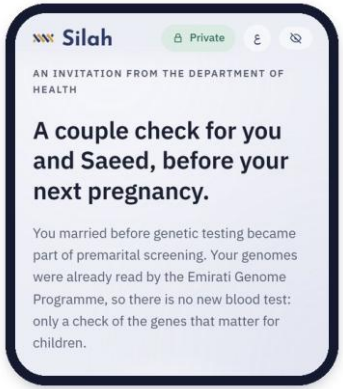
Architecture or workflow diagram*

# One phone app, five services, only signed or blinded messages



Target users and intended outcome*

# Three moments where knowing the risk still changes an outcome




## Before the next pregnancy

Couples married before the mandate, invited to a couple check on genomes already held.

Outcome: an at-risk couple knows before the next pregnancy and reaches a counsellor.




## At birth

Any at-risk couple expecting a baby, whatever they chose before.

Outcome: the known change is tested in the first days, ahead of the 21-day newborn genome report, so SMA or Krabbe is treated before signs.




## In the family

Relatives of every carrier, invited without saying who sent the invitation.

Outcome: relatives come to testing early, with a reason, and nobody is exposed.

**15,628**
Emirati births in Abu Dhabi in 2024

**3 in 4**
or more of 2026 Emirati births are to couples the genetic test never reached (estimate)

**900,000+**
Emirati genomes already sequenced

**565 / 11,748**
couples found incompatible since 2022

How will success be measured?*

# Measured on the DoH's KPI and on understanding, never on choices

| Measure | How it is counted |
|---|---|
| Non-compatible couples referred to counselling | DoH KPI, target 90%; live in the pathway view |
| Days from result to counselling | Median, by language |
| Understanding: the four teach-back points | Asked by a researcher at the visit; Silah vs the current report, randomised by week |
| "Compatible means zero risk" misconception | Among compatible couples, both arms |
| Married couples who accept the couple check | Per invitation sent, by declared signal |
| At-risk pregnancies with a birth plan | Newborns tested in days 1–3, from maternity records |
| Relatives tested per invitation | Redeemed tokens; anonymous by design |
| Equity | Every measure by language and reading confidence; counts under 5 suppressed |

### Not a success measure

Marriages cancelled, or any single reproductive choice. Choices are reported, never ranked.

### Already tracked in the prototype

Referral KPI, pathway funnel, teach-back shared, birth plans active, invitations redeemed, consent withdrawn.

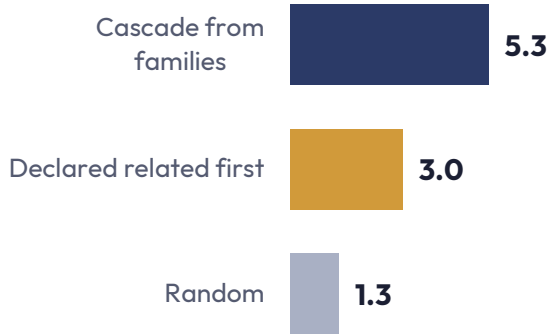
Evidence supporting the approach*

# Exact arithmetic, privacy held end to end, outreach order tested

## Tested in the prototype

- 158 unit tests. The kernel matches values worked out by hand: Fatima's chance is 237/62,468, about 1 in 260, where the certificate says "no risk".
- 23-check story through the real interface: neither partner's private finding reaches the other or the clinic.
- 0 WCAG 2.2 AA violations on 18 screens, both languages.
- Offline on an Android 13 emulator in airplane mode.
- Readable: English at US grade 6.9, against 14.2 for the DoH sheet couples sign today; Arabic 84.1 against 67.4 on OSMAN (higher is easier).

## Which couples to invite first




At-risk couples found per 100 tests, after 200 tests. Simulated consanguineous population, 5 seeds, published Emirati carrier frequencies for 7 genes.

## Safe answers, checked

| 52 questions | Grounded | Traps | Steers |
|---|---|---|---|
| Gemma E4B | 22/22 | 28/30 | 0 |
| Gemma E2B | 21/22 | 30/30 | 0 |
| Qwen 4B | 22/22 | 30/30 | 1 |

22 grounded questions and 30 traps, Arabic and English. Steers: directive sentences the model wrote, before the verifier. A chance moved to the wrong outcome is blocked: 84 of 84 mutants.

- Early SMA treatment: 22 of 25 babies treated before signs later walked (De Vivo 2019).
- Official compatible-couple wording: "No risk for either the other partner or children to be carrier or diseased" (DoH Standard V3, 2024).

Data sources and legal right to use them*

# No personal data: public standards, published figures, open licences

| Source | Owner | Used for | Right to use | Personal data |
|---|---|---|---|---|
| Premarital Screening Standard V3 (2024) | DoH Abu Dhabi | Panel (570 printed names = 563 genes), report wording, 90% KPI | Public standard, cited | No |
| HGNC gene symbols | HUGO Gene Nomenclature Committee | Mapping the panel to current symbols | CC0 | No |
| Jain et al. 2025; Rabea et al. 2024 | Authors / publishers | Emirati carrier frequencies (7 genes) | Published facts, cited | Aggregate |
| NURTURE, SPR1NT and other trials | Authors / journals | Birth-plan statements | Published facts, cited | Aggregate |
| Programme figures (news reports) | Gulf News, Aletihad, Gulf Today | 565 of 11,748; 900,000 genomes; births | Published facts, cited | No |
| Gemma 4 E2B / E4B | Google | Optional answers on the device | Apache-2.0 | No |
| Neural voices (ar-AE, en-GB) | Microsoft | Pre-recorded narration | Prototype use | No |
| Family, genomes, clinic records | Written for this prototype | The whole demonstration | Own work | Fictional |

In a deployment, genomic data stays with the lab and on the person's phone, and services are hosted in the UAE (Federal Law No. 2 of 2019). Couple results from EGP genomes need fresh consent, which Silah collects.

Risks and safeguards*

# Every risk has a safeguard in code, and a person is one tap away

| Risk | What could go wrong | Safeguard in Silah |
|---|---|---|
| Privacy | A partner or relative learns a finding | Each partner sees only shared genes; sharing is by choice, signed, revocable; invitations name no sender |
| Security | A forged result or tampered consent | Lab results Ed25519-signed and checked on the phone; receipts hash-chained; vault encrypted |
| Bias | Outreach that profiles families | Ranking uses declared signals only, never genetics; every measure by language; counts under 5 suppressed |
| Explainability | Numbers nobody can check | Every number is an exact fraction with its arithmetic and source; assumptions labelled |
| Misuse | Steering a decision, or a wrong number | Verifier drops directive or certain sentences, and any chance given for the wrong outcome; options at equal weight |
| Human oversight | Distress, religion, termination | Hard routes to a person before any model; counsellor brief; quick hide |
| Failure modes | Model fails or omits a risk | Common risk questions answered from fact cards; model optional; works offline |

Hardware requirements

# No hardware to buy: Silah runs on the phones people already have

## Phone

Any recent browser. Installs as an app, works offline, about 7 MB stored for offline use. Arabic and English, larger text.

## Optional model on the phone

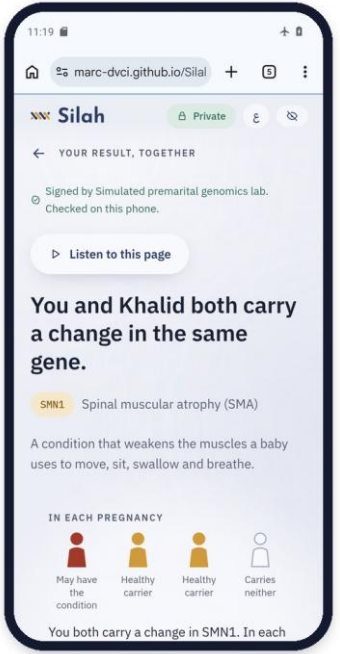
Gemma 4 E2B in the browser on WebGPU, a one-time download of about 3 GB. Without it, common questions still work.

## Counsellor computer

Gemma 4 E4B through llama.cpp. Measured on an RTX 4070: median 1.8 s per answer.

## Health-system services

Standard servers hosted in the UAE. No sensor, wearable or device touches the body, so there is no device safety risk.




Android 13, airplane mode: the live app reopened offline

Path to pilot and scale*

# A pilot at one clinic, then a priced path to every married couple

| Stakeholder | Role |
|---|---|
| DoH Abu Dhabi | Owns the programme; approves the check, outreach and result format |
| M42 · EGP | Holds the genomes, runs the couple check, signs results |
| SEHA · Corniche Hospital | Counselling clinic, maternity, IVF + PGT |
| Premarital centres | Issue the signed result with the certificate |

**Regulation:** DoH Policy on AI in Healthcare (the model only rephrases checked facts; no decision depends on it); UAE health data law, data in the UAE.

**Who pays:** the programme owner, DoH or M42. Families never pay. No new sequencing, and no server cost per question: the model runs on the device. Timeline is an assumption.

**Months 0–3** DoH approval of the result format and consent; UAE hosting; clinical review of the birth plan

**Months 3–9** One centre, Corniche counselling and maternity; Silah vs current report, randomised by week

**Months 9–18** Invite couples married before the mandate; birth plans across SEHA maternity units

## Why DoH would pay

**Measured:** 9 genetic counsellors and 8 clinical geneticists for nearly 11 million people.

**Assumed:** 20,000 married couples re-checked a year find about 960 at risk, 8 times today's flow.

**Priced:** 20,000 re-checks plus about 2,600 premarital couples, at an assumed AED 100 each: about AED 2.26 million a year. Silah does the explaining and triage so those 17 people can counsel.

Pre-existing IP and third-party materials*

# Written for this challenge; every third-party part is open-licensed

## Silah's own work

Risk kernel, verifier, consent receipts, invitations, phone app, clinic side, fact cards, Arabic and English text, evaluation set and tests: written by Marc Donovici for this challenge, October–November 2026.

Public at github.com/Marc-Dvci/Silah under the Apache License 2.0.

## Restrictions

No proprietary data. Narration voices need a commercial licence for production.

| Component | Licence |
|---|---|
| Preact, Vite, Workbox (vite-plugin-pwa) | MIT |
| @noble/curves, @noble/hashes | MIT |
| @cloudflare/blindrsa-ts (RFC 9474) | Apache-2.0 |
| Transformers.js; llama.cpp | Apache-2.0; MIT |
| Gemma 4 E2B / E4B weights | Apache-2.0 |
| qrcode | MIT |
| IBM Plex Sans, Sans Arabic, Mono; Reem Kufi | SIL Open Font Licence |
| HGNC gene symbols | CC0 |
| Microsoft neural voices (narration) | Prototype use |

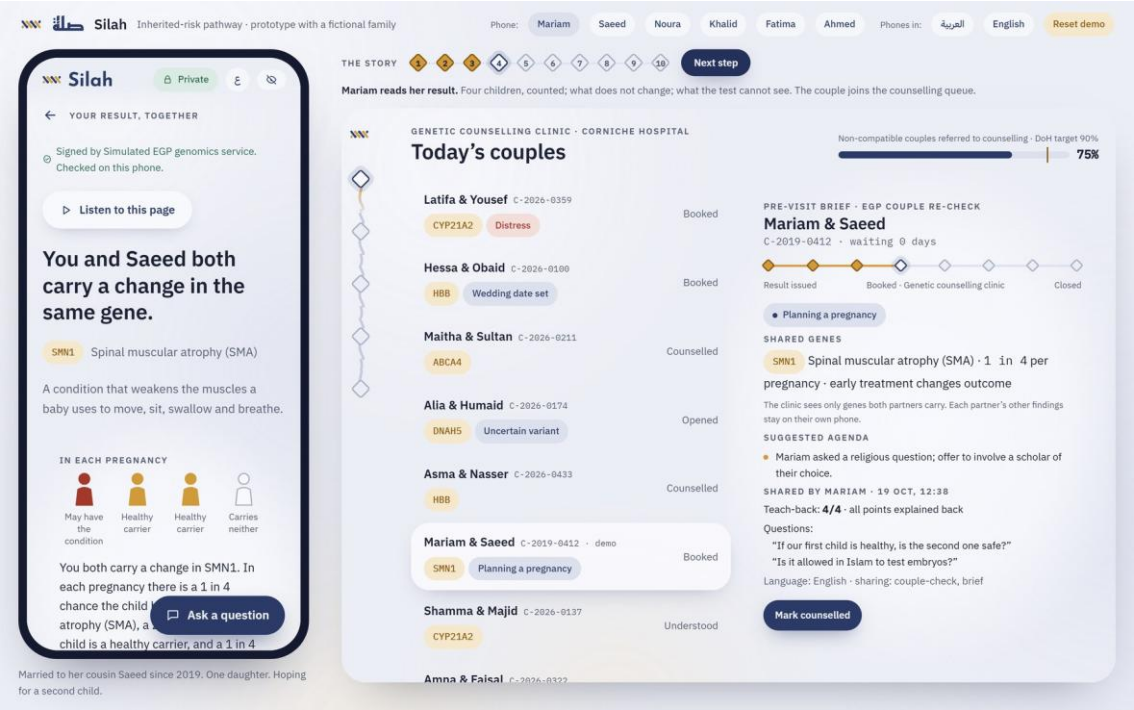
Conflict of interest declaration*

# None. No member of the team has a conflict of interest to declare.

| | |
|---|---|
| **No** | Employed by, or contracted to, the Department of Health – Abu Dhabi |
| **No** | Employed by, or contracted to, Future Health or a challenge collaborator (Microsoft, Inception42, MBZUAI, Hack2skill) |
| **No** | Family or close personal relationship with a judge, organiser or collaborator staff member |
| **No** | Financial interest in an organisation involved in running or judging the challenge |
| **No** | Access to non-public information about the challenge or its evaluation |

Additional Slide

# Try it: phone and health system side by side, in one browser




- Ten steps with a Next button; each step moves both screens and finishes itself.
- One switch puts every phone in Arabic or English.
- Fictional family: Mariam and Saeed (married 2019), Noura and Khalid (expecting), Fatima and Ahmed (engaged, "compatible").
- Runs offline after the first visit; no account, no server, no API key.
- [marc-dvci.github.io/Silah](marc-dvci.github.io/Silah)

Scan for the product page, the live demo, the film and the documents.

Additional Slide

# Assumptions, dependencies and limitations, each shown where it applies

## Assumptions

- Carrier frequency of 1 in 200 for panel genes without a published Emirati figure; 7 genes use published figures.
- The panel finds 90% of a gene's disease-causing changes (99% for a known family change), shown beside every number that uses it. From 80% to 99%, Fatima's 1 in 260 moves from 1 in 140 to 1 in 1,300.
- Re-check volume (20,000 couples a year), price (AED 100 per couple) and the pilot timeline.

## Dependencies

- A DoH decision, with fresh consent from both partners, to return couple results from genomes EGP already holds. Silah provides the consent flow.
- A lab-signed result issued with the existing digital certificate; the integration with DoH, M42 and Malaffi is specified in the repository.
- Gemma 4 (Apache-2.0) for optional answers; common questions need no model.

## Limitations of the prototype

- The family, their genomes and every clinic record are fictional; the lab, clinic, hospital and invitation service run in the browser.
- No clinician or patient has reviewed Silah yet, and understanding is measured on the text only. A review of all 68 statements and a reader check are set out.
- Model answers: 52 questions per model, every transcript published. The verifier checks each number and the outcome it belongs to, not every medical statement.

Full list with sources: docs/ASSUMPTIONS_AND_LIMITATIONS.md at **github.com/Marc-Dvci/Silah**

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# Future Health Digital Innovation Challenge

From signals to solutions.

# Thank you

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